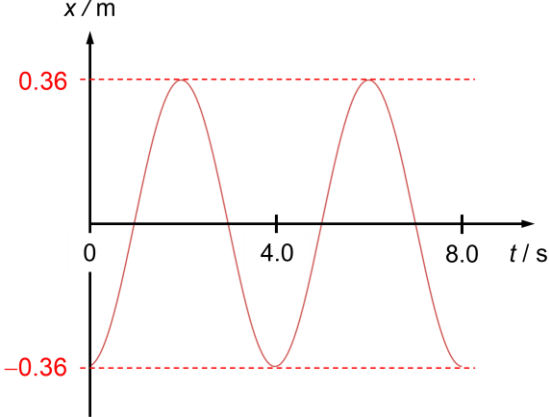


Qn	Suggested Solutions
4(a)	$mg = kx_0$
(b)	Take the downward direction as positive. Resultant force with extension below equilibrium point, $F_R = mg - k(x_0 + x)$ where x_0 is the extension at equilibrium.
	By Newton's second law, $mg - k(x_0 + x) = ma$ $kx_0 - kx_0 - kx = ma$ since $mg = kx_0$ $a = -\frac{k}{m}x$ (Shown)
	OR
	Define upwards as positive $-mg + k(x_0 + x) = ma$
	$-kx_0 + kx_0 + kx = ma$ $a = \frac{k}{m}x$ Since a and x are opposite in direction, $a = -\frac{k}{m}x$
(c)(i)	From graph, $T = 4.0$ s $f = \frac{1}{T} = \frac{1}{4.0}$
	$f = 0.25$ Hz
(ii)	Comparing $a = -\omega^2 x$ and $a = -\frac{k}{m}x$, $\omega^2 = \frac{k}{m}$ $4\pi^2 (0.25)^2 = \frac{28}{m}$
	$m = 11.3$ kg
(iii)	Max $E_k = \frac{1}{2}mv_o^2$ $1.8 = \frac{1}{2}(11.3)v_o^2$ $v_o = 0.564$ $\approx 0.56 \text{ m s}^{-1}$

(iv)	$v_0 = \omega x_0$ $x_0 = \frac{v_0}{\omega}$ $= \frac{0.564}{2\pi(0.25)}$ $= 0.359$ $\approx 0.36 \text{ m}$
(d)	

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H2 (9749) Physics

2025 H2 Preliminary Exam Paper 3 Guide